



PILLAI COLLEGE OF ENGINEERING

PROJECT REPORT

LiFi-Based Chat System

**Submitted in partial fulfilment of the requirements for the degree of
Bachelor of Technology**

In

ELECTRONICS AND TELECOMMUNICATION ENGINEERING

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1. Introduction

LiFi is a wireless communication technology that uses light to transmit information. In visible light communication, an LED can vary its optical intensity at a speed much faster than the human eye can observe. These intensity variations can therefore represent digital data.

The LiFi-Based Chat System developed in this project demonstrates the complete digital communication process in software. A text message is represented as numerical bytes, converted into binary information, encoded, modulated using OOK, transmitted through a simulated optical channel, received by a simulated photodetector, demodulated, decoded and displayed at the receiver.

1.1 Problem Statement

The objective is to demonstrate how a text message can be transmitted using basic optical wireless communication principles without requiring physical hardware. The system must represent the message digitally, apply suitable modulation, simulate the channel and receiver, and recover the transmitted information correctly.

1.2 Scope

- End-to-end software simulation of a LiFi communication link.
- Text/byte-to-binary conversion.
- Encoding and decoding stages.
- On-Off Keying modulation and demodulation.
- LED/LiFi transmitter and photodetector simulation.
- Simplified optical channel modeling.
- Signal observation using Simulink Scope blocks.

1.3 Importance of LiFi Technology

LiFi technology is an emerging form of wireless communication that uses visible light for transmitting information. Unlike conventional radio-frequency communication systems, LiFi uses variations in light intensity to represent digital data. Since light sources such as LEDs can switch at extremely high speeds, the modulation is generally not visible to the human eye while still allowing information to be transmitted.

LiFi has gained significant attention because of its potential for high-speed communication, reduced radio-frequency interference, and improved communication in environments where conventional wireless technologies may not be suitable. The technology can be applied in indoor communication systems, smart environments, hospitals, aircraft, underwater communication, and other specialized applications.

2. Objectives and Scope

2.1 Objectives

- To study the working principle of LiFi and Visible Light Communication.
- To develop a software-based chat communication model using MATLAB and Simulink.
- To convert a text message into digital byte and binary data.
- To implement encoding before transmission and decoding after reception.
- To use OOK as the optical modulation technique.
- To simulate an LED/LiFi transmitter and optical channel.
- To simulate photodetector reception.
- To recover and display the transmitted information.
- To demonstrate a complete digital communication chain in a graphical Simulink model.

2.2 Project Scope

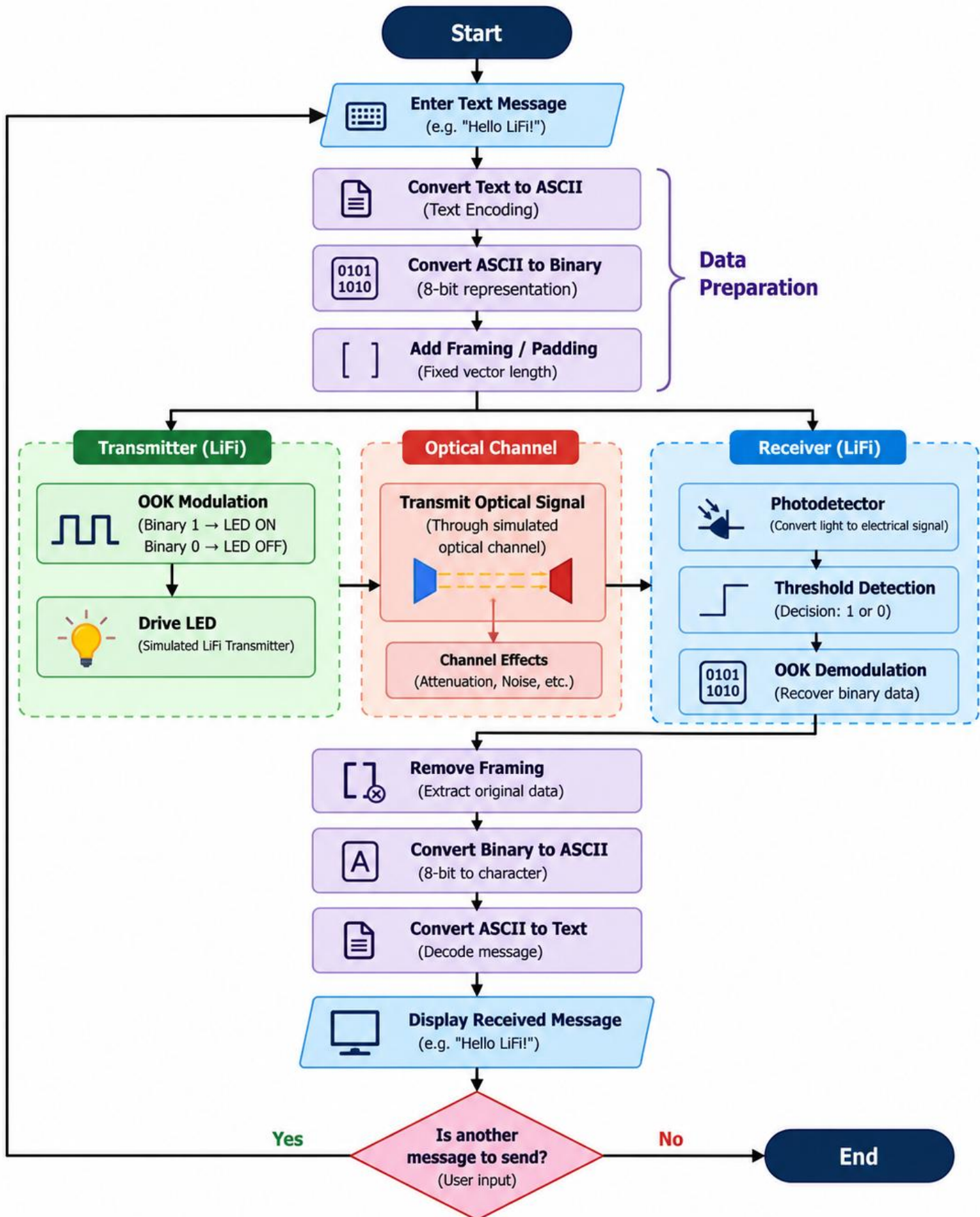
The project focuses on a functional block-level simulation of a LiFi communication system. All transmitter and receiver operations are modeled mathematically or through signal-processing blocks. The current implementation uses the sample message "Hello LiFi!" represented as uint8 byte values and padded with zeros to maintain a fixed vector length suitable for Simulink processing.

The project also includes graphical visualization of the optical signal and detected bits, along with a performance analysis section displaying important parameters such as total transmitted bits, correctly received bits, erroneous bits, and BER.

This project is intended primarily for **academic demonstration and learning purposes**. It provides a clear understanding of the complete digital communication process, including:

Message Input → Text Encoding → Binary Conversion → Framing → OOK Modulation → LED/LiFi Transmission → Optical Channel → Photodetection → Threshold Detection → Demodulation → Decoding → Binary-to-Text Conversion → Received Message

2.3 Flowchart



3. System Design and Methodology

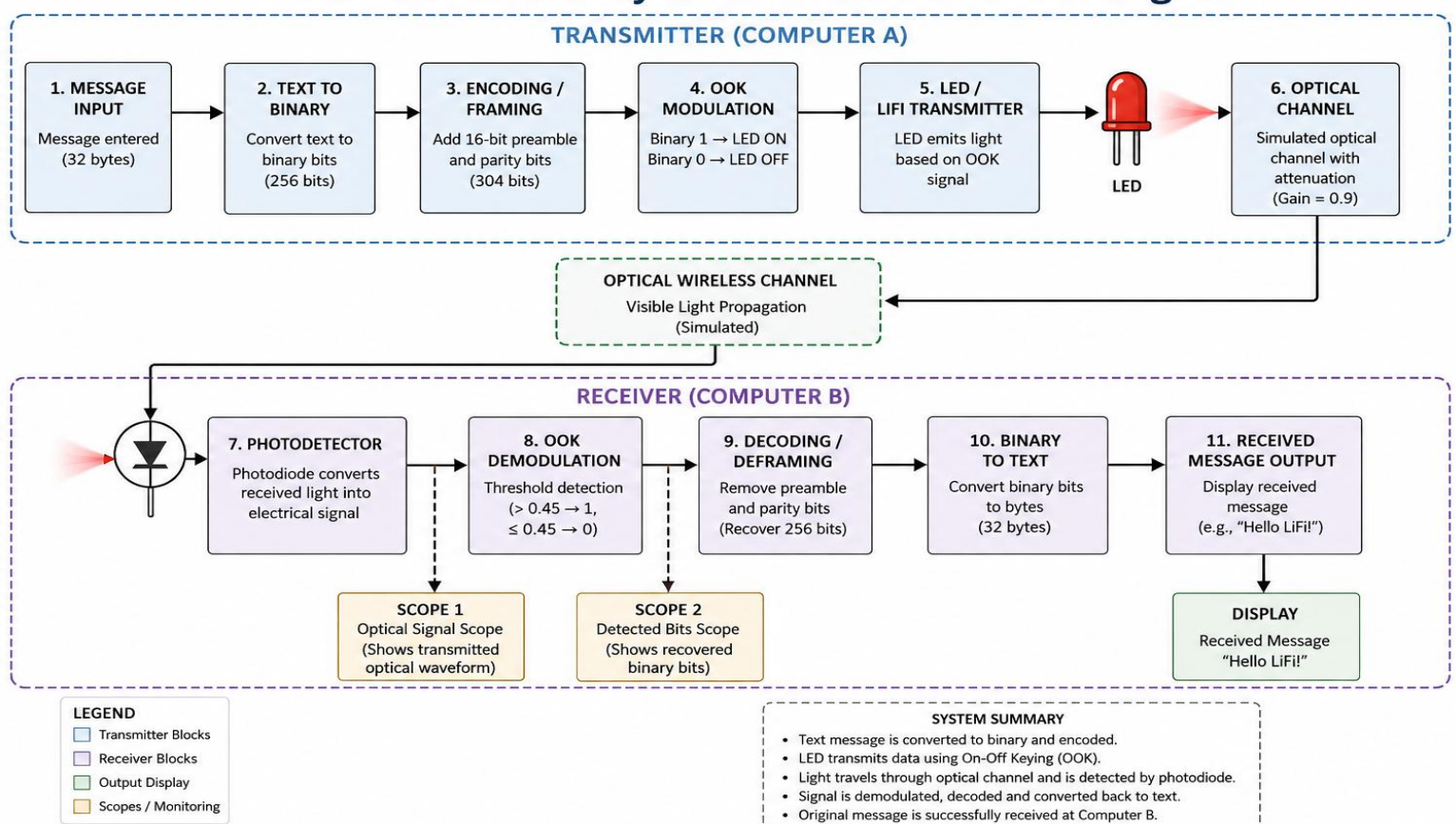
The system is divided into transmitter, optical channel and receiver sections. The complete processing sequence is:

1. Message Input
2. Text to Binary
3. Encoding
4. OOK Modulation
5. LED / LiFi Transmitter
6. Optical Channel
7. Photodetector
8. OOK Demodulation
9. Decoding
10. Binary to Text
11. Received Message Output

3.1 Block Diagram Description

Message Input stores the sample message as uint8 bytes. Text to Binary converts each byte into an 8-bit representation. Encoding prepares the transmission frame. OOK Modulation maps binary information to ON/OFF optical levels. The LED/LiFi transmitter represents the optical source. The Optical Channel models propagation and attenuation. The Photodetector represents optical-to-electrical conversion.

LiFi-Based Chat System – Simulink Block Diagram



3.2 Processing Sequence

1. The model automatically initializes the message input variable through the Simulink InitFcn callback.
2. The message is supplied as a uint8 byte vector.
3. Each byte is converted into binary data.
4. The binary information passes through encoding.
5. OOK maps binary 1 and 0 to optical ON and OFF levels.
6. The signal passes through the LED/LiFi transmitter and optical channel.
7. The photodetector receives the optical signal.
8. OOK demodulation recovers the digital information.
9. Decoding and Binary to Text reconstruct the received bytes.

3.3 Automatic Model Initialization

Variables stored only in the MATLAB base workspace are removed after MATLAB is restarted. To make the model self-contained, the input variable is automatically recreated using the model InitFcn callback.

```
assignin('base','msgInputBytes', uint8([72 101 108 108 111  
32 76 105 70 105 33 zeros(1,21)]));
```

.This automatic initialization provides several important advantages:

- The required input data is automatically available whenever the model is opened.
- The user does not need to manually create variables in the MATLAB Command Window.
- The simulation can be executed even after MATLAB has been closed and restarted.
- The model becomes more reliable and easier to demonstrate.
- The possibility of workspace-related errors is reduced.
- The project becomes more suitable for academic submission because it can run independently without manual preparation.

The use of the **InitFcn callback** (initialization function) therefore improves the portability and usability of the simulation. Whenever the LiFi Simulink model is initialized, the required message data is recreated automatically before the simulation begins.

This ensures that the complete communication process can be demonstrated repeatedly without requiring the user to manually configure the MATLAB workspace each time.

4. Theory and Mathematical Model

4.1 LiFi and Visible Light Communication

Visible Light Communication uses optical intensity variations to carry information. In a basic LiFi link, an LED acts as the transmitter and a photodetector acts as the receiver. Rapid changes in light intensity can represent binary data.

4.2 Digital Representation of Text

The sample message "Hello LiFi!" is represented by the decimal byte sequence:

72, 101, 108, 108, 111, 32, 76, 105, 70, 105, 33

Each byte contains 8 bits. Therefore, fixed 8-bit character representation requires Total bits = $8 \times N$.

4.3 On-Off Keying (OOK)

OOK is a simple digital modulation technique suitable for intensity-modulated optical communication. Binary 1 is represented by optical light ON with amplitude A, while binary 0 is represented by light OFF with amplitude 0.

$$s(t) = A \text{ for bit 1; } s(t) = 0 \text{ for bit 0}$$

4.4 Optical Channel

A simplified optical channel is represented by a gain or attenuation factor h:

$$r(t) = h \times x(t)$$

4.5 Photodetection and Threshold Detection

A photodetector converts incident optical intensity into an electrical signal. A simplified normalized receiver can preserve the received signal level while representing optical-to-electrical conversion. The receiver then compares the signal with a decision threshold T.

$$\text{Detected bit} = 1 \text{ if } y(t) \geq T; \quad \text{Detected bit} = 0 \text{ if } y(t) < T$$

4.6 Bit Error Rate

Bit Error Rate is defined as $BER = \text{Number of erroneous bits} / \text{Total transmitted bits}$. For an ideal or low-loss demonstration, the expected BER is approximately zero when the received data matches the transmitted data.

5. MATLAB and Simulink Implementation Details

The project is implemented using MATLAB and Simulink. MATLAB supports variable preparation and model initialization, while Simulink provides the graphical block-diagram environment used to construct and execute the communication chain.

5.1 Main Blocks and Functions

Block	Type	Function
Message Input	Constant	Value = msgInputBytes
Text to Binary	MATLAB Function	Converts uint8 bytes to bits
Encoding	MATLAB Function	Transmission frame processing
OOK Modulation	MATLAB Function	Maps bits to OOK levels
LED/LiFi Transmitter	Gain	Optical transmitter representation
Optical Channel	Gain	Example attenuation gain 0.9
Photodetector	Gain	Optical-to-electrical representation
OOK Demodulation	MATLAB Function	Recovers binary information
Decoding	MATLAB Function	Recovers transmitted frame
Binary to Text	MATLAB Function	Converts bits to bytes
Received Output	Display	Displays received information
Scope	Scope	Observes signals

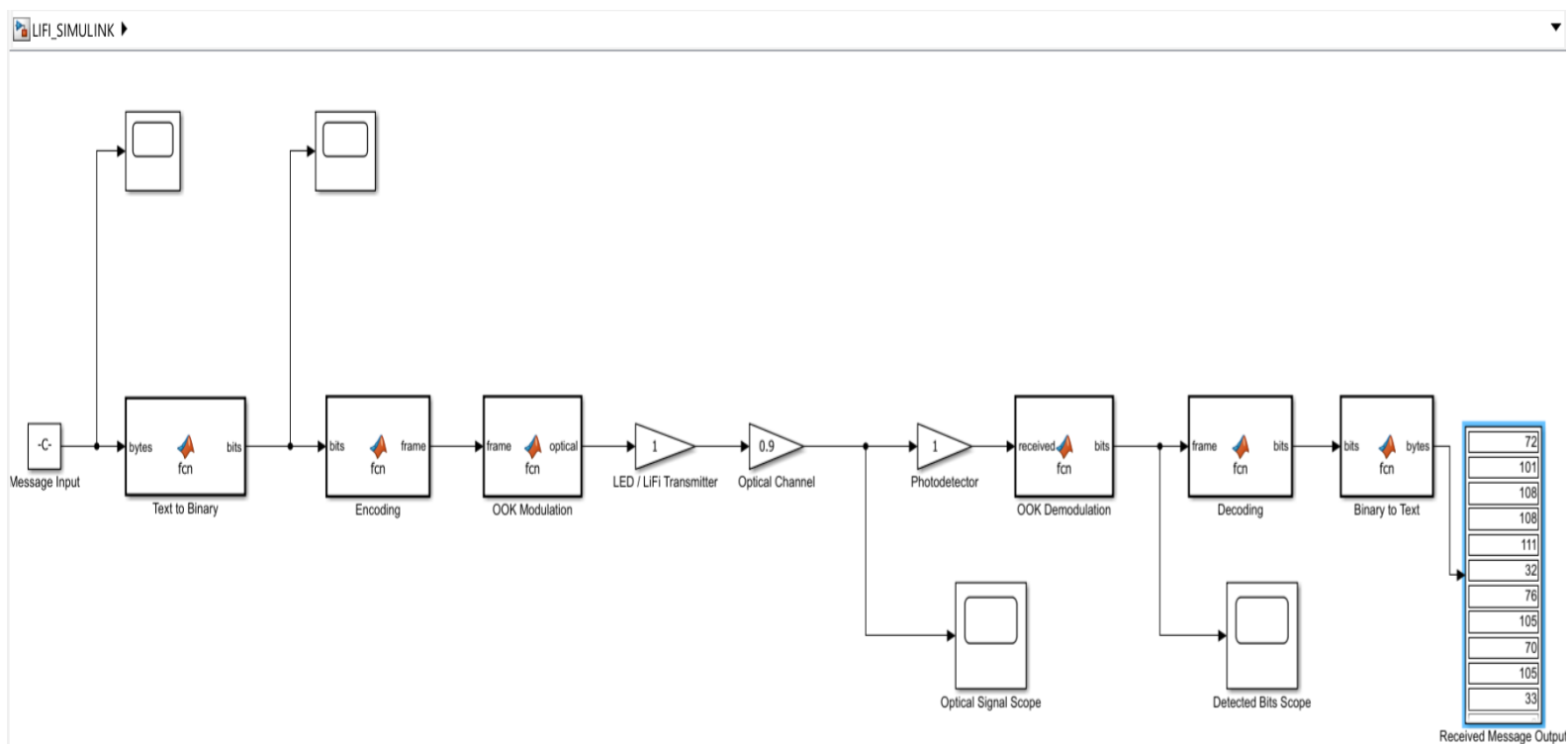
5.2 Model Configuration

- Simulation environment: MATLAB Simulink.
- Message data type: uint8.
- Sample message: "Hello LiFi!".
- Fixed message vector: 32 bytes with zero padding.
- Modulation: On-Off Keying (OOK).
- Optical channel: simplified gain/attenuation model.
- Receiver: photodetector followed by OOK demodulation and decoding.
- Simulation: fixed-step discrete configuration as used in the model.

5.3 Connection Sequence

Message Input → Text to Binary → Encoding → OOK Modulation → LED/LiFi Transmitter → Optical Channel → Photodetector → OOK Demodulation → Decoding → Binary to Text → Received Message Output

Scope blocks are connected as branches from the main signal path so that signals can be observed without interrupting the communication chain.



6. Modulation and Detection Technique

6.1 On-Off Keying (OOK) Modulation

The modulation technique used in this project is **On-Off Keying (OOK)**. OOK is one of the simplest modulation techniques used in optical and digital communication systems. In On-Off Keying, binary information is represented using two different states of the optical signal. When the transmitted binary data is **1**, the LED is considered to be switched ON and optical light is transmitted. When the binary data is **0**, the LED is switched OFF and no optical signal is transmitted.

Therefore, the basic representation used in this project is:

Binary 1 → Light ON

Binary 0 → Light OFF

This technique is suitable for a LiFi communication system because information can be transmitted by rapidly changing the intensity of light. These changes occur at a speed that can represent digital information while the basic concept remains similar to wireless data communication.

Mathematically, the OOK transmitted signal can be represented as:

$s(t) = A$, when the transmitted bit is 1

$s(t) = 0$, when the transmitted bit is 0

6.2 OOK Demodulation and Detection

At the receiver side, the optical signal is converted back into an electrical signal using a simulated photodetector. The received signal may have reduced amplitude because of the optical channel characteristics.

The receiver uses a **threshold detection method** to determine whether the received signal represents binary 1 or binary 0.

The decision process is:

If the received signal is greater than or equal to the threshold, the detected bit is 1.

If the received signal is less than the threshold, the detected bit is 0.

In mathematical form:

Detected bit = 1, if $y(t) \geq T$

Detected bit = 0, if $y(t) < T$

where **$y(t)$** is the received signal and **T** is the decision threshold.

Advantages of OOK Modulation

OOK modulation has several advantages for a basic LiFi communication system:

- It is simple to understand and implement.
- It directly represents binary data using ON and OFF states.
- It requires relatively simple transmitter and receiver processing.
- It is suitable for demonstrating optical communication principles.

7. Optical Channel and Photodetection

7.1 Simulated Optical Communication Channel

The optical channel represents the communication path between the LiFi transmitter and receiver.

In a real-world LiFi system, the optical signal may be affected by several factors, including:

- Distance between transmitter and receiver.
- Light attenuation.
- Ambient light.
- Obstacles in the communication path.
- Reflection from surrounding surfaces.
- Alignment between the LED and photodetector.

For the purpose of this project, the optical channel is simplified and represented mathematically using a channel gain.

7.2 Photodetection

At the receiver side, a **photodetector** is used to receive the incoming optical signal.

In an actual LiFi communication system, a photodetector converts incident light energy into an electrical signal. Common devices used for this purpose include photodiodes.

The basic conversion process is:

Optical Signal → Electrical Signal

The electrical signal produced by the photodetector can then be processed by the receiver.

In the developed simulation, the photodetection process is represented mathematically rather than through physical hardware.

The simulated photodetector performs the same conceptual function as a real optical receiver. It receives the optical signal and provides an electrical representation that can be processed by the OOK demodulation and threshold detection stages.

7.3 Received Optical Signal

The received optical signal may differ from the transmitted signal because of channel effects.

The project monitor displays the received waveform under:

RECEIVED OPTICAL SIGNAL

This graph allows the transmitted and received signals to be compared.

The received signal is then processed by the threshold detector. As long as the received amplitude for a transmitted binary 1 remains above the decision threshold, the receiver can correctly detect the transmitted bit.

8. Performance Analysis and Bit Error Rate

8.1 Importance of Performance Analysis

The performance of a digital communication system is determined by how accurately the receiver can recover the transmitted information.

One of the most important performance parameters is the **Bit Error Rate (BER)**.

BER measures the number of incorrectly received bits compared with the total number of transmitted bits.

The formula for BER is:

$$\text{BER} = \text{Number of Erroneous Bits} / \text{Total Number of Transmitted Bits}$$

A lower BER indicates better communication -performance.

An ideal communication system would have:

$$\text{BER} = 0$$

This means that every transmitted bit was received correctly.

8.2 BER Analysis in the Developed Project

Bit Error Rate (BER) analysis is used to evaluate the accuracy and reliability of the developed LiFi communication system. BER represents the ratio of incorrectly received bits to the total number of transmitted bits. It is an important performance parameter because it shows whether the transmitted information is successfully recovered at the receiver.

In the developed project, the original message is first converted into binary data and transmitted through the simulated LiFi communication channel using On-Off Keying (OOK) modulation. At the receiver side, the optical signal is detected, demodulated, and converted back into binary information.

8.3 Interpretation of BER Results

A BER value of zero indicates that:

- The modulation process worked correctly.
- The optical channel maintained sufficient signal strength.
- The photodetection process correctly received the signal.
- The threshold detector correctly identified binary 1 and binary 0.
- The demodulation process successfully recovered the digital information.
- The final received message matched the original transmitted message.

The BER result therefore provides quantitative confirmation that the developed communication system is functioning correctly under the selected simulation conditions.

In practical communication systems, BER may increase because of noise, interference, attenuation, or receiver limitations. Therefore, BER analysis is an important method for evaluating communication system reliability.

9. Results and Discussion

9.1 Test Message

"Hello LiFi!"

The important decimal byte values corresponding to the test message are:

72 101 108 108 111 32 76 105 70 105 33

The remaining entries are zeros used as padding to maintain a fixed 32-byte vector size.

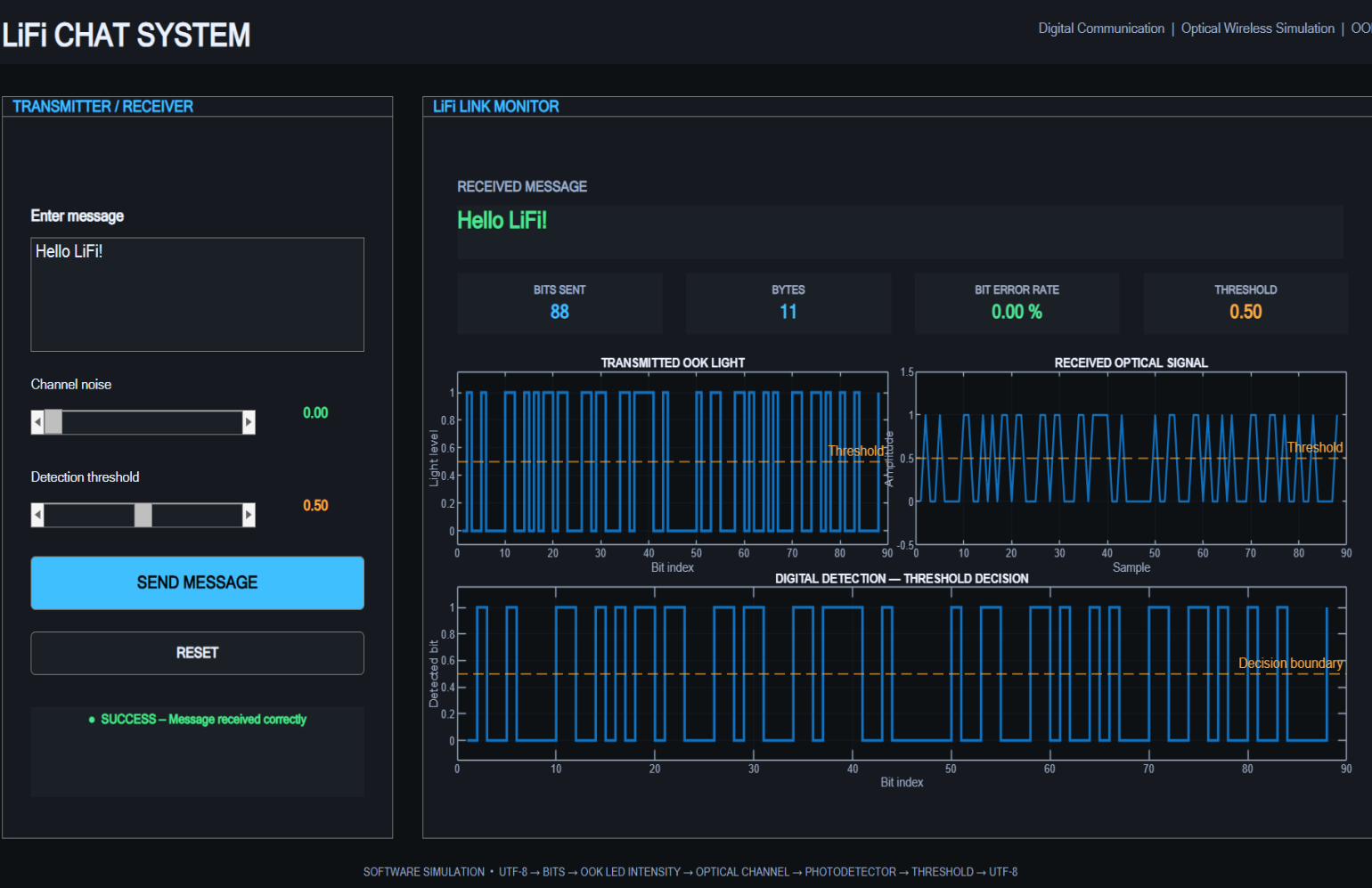
9.2 Result Interpretation

A successful simulation run without Simulink block errors demonstrates that the complete transmitter-to-receiver signal path is operational. When the received byte sequence matches the transmitted byte sequence, the communication chain has preserved the transmitted information correctly.

9.3 Representative Output

The received values 72, 101, 108, 108, 111, 32, 76, 105, 70, 105 and 33 correspond to:

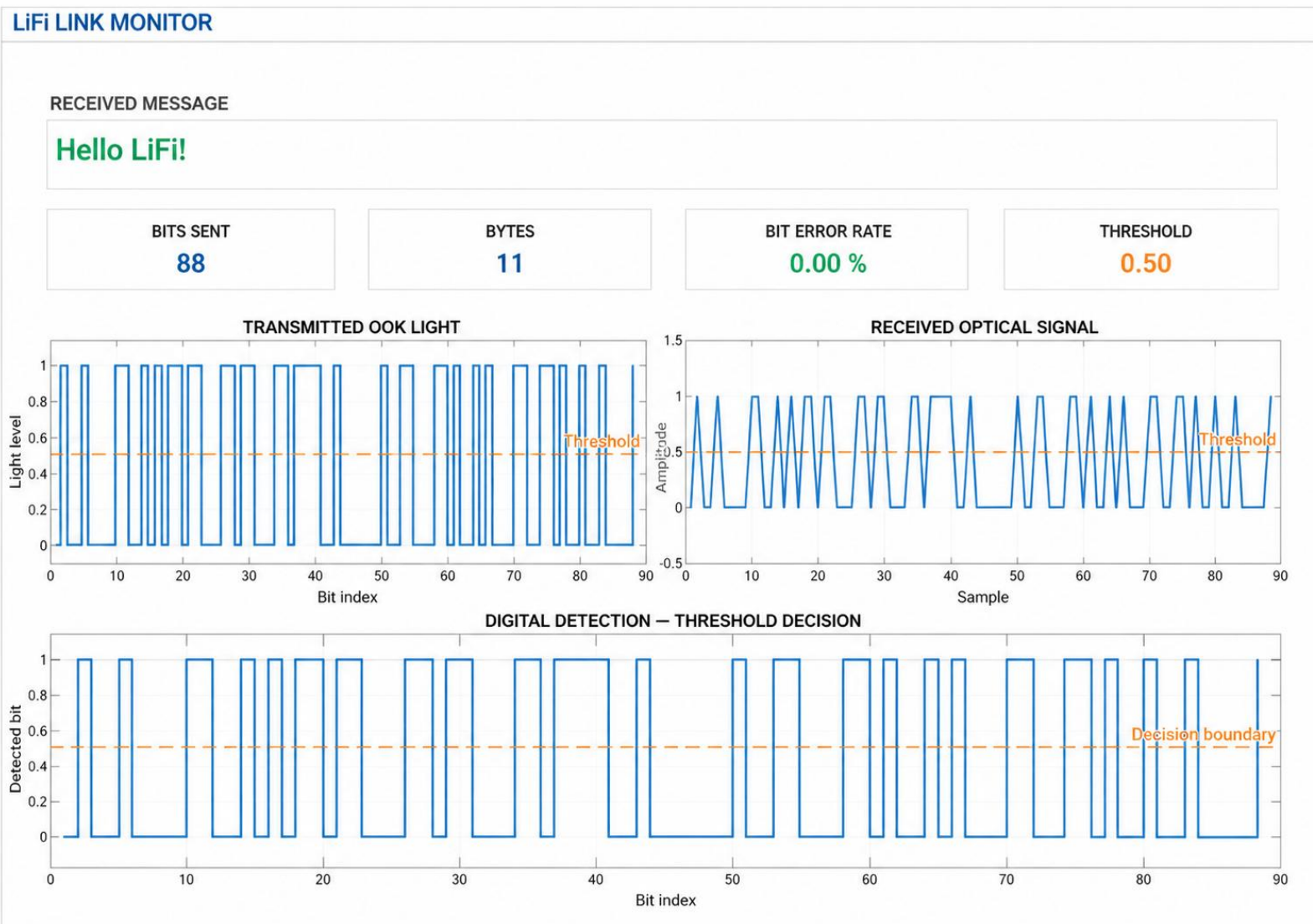
H e l l o L i F i !



9.4 Discussion and Observations

The simulation demonstrates the fundamental principle of LiFi communication: digital information can be represented by changes in optical intensity and recovered at the receiver. OOK provides a direct and easy-to-understand mapping between binary data and light ON/OFF states.

- The complete model runs as a software-only simulation.
- The message is represented as a digital byte sequence before transmission.
- OOK provides clear optical ON/OFF states.
- The optical channel and photodetector provide conceptual models of a practical LiFi link.
- The receiver can recover the transmitted information under the configured simulation conditions.
- The InitFcn callback allows the project to run again after MATLAB is restarted.



10. Conclusion and Future Scope

10.1 Conclusion

The LiFi-Based Chat System Using MATLAB and Simulink demonstrates a complete software model of digital text communication through a simulated optical wireless link. The project includes message input, binary conversion, encoding, OOK modulation, LED/LiFi transmission, optical channel propagation, photodetection, demodulation, decoding and received output. The project provides an effective academic demonstration of how LiFi principles can be connected with digital communication theory without requiring physical hardware. Automatic initialization through the Simulink InitFcn callback also improves reliability when the model is reopened.

10.2 Limitations

- No physical LED or photodetector is used.
- The optical channel is simplified.
- Detailed ambient light and multipath effects are not included.
- Advanced channel coding and error correction are not implemented.
- The demonstration uses a fixed-size message vector for stable signal dimensions.

10.3 Future Scope

- Add interactive text input directly connected to the simulation.
- Add adjustable optical noise and ambient-light interference.
- Calculate BER for different channel conditions.
- Add threshold control and performance analysis.
- Implement Hamming or other error-control coding.x
- Compare OOK with other optical modulation techniques.

11. References

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